

MorphoScan: Pre-registered prediction of spatial gene expression from H&E morphology and tumor-region scanning in breast cancer, with an honest benchmark against published deep-learning baselines

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Abstract

Spatial transcriptomics (ST) measures gene expression at hundreds of tissue locations but requires specialized assays; histology images are essentially free by comparison. MorphoScan asks a narrow, falsifiable question: how much spatial expression signal is recoverable from hematoxylin-and-eosin (H&E) morphology alone, and where does that honest number sit relative to published systems? Every gate was pre-registered before results were seen (PREREG.md and PREREG-2.md, committed first). Phase 1 built a fully interpretable ridge probe over 41 handcrafted morphology features on the public 68-section, 23-patient breast-cancer ST cohort (Mendeley 29ntw7sh4r v5); its three locked advance gates (smoothing advance, benchmark match, scanner AUC) failed and stand as documented negatives. Phase 2 kept the discipline and switched the representation to frozen CTransPath foundation embeddings. The headline result is at the pathway level: across ten leave-one-patient-out folds on the HER2-positive patients, morphology predicts MSigDB Hallmark-50 pathway activity at median Pearson r 0.326 [0.104, 0.571] (patient-level bootstrap), permutation $p = 0.0011$ at 10,000 permutations. Per-gene prediction is weaker (median r 0.046 [0.036, 0.072]) and does not match ST-Net's published 0.19; that gap is logged as measured. Spatial smoothing adds a small but real gain (median Δ 0.0061 [0.0046, 0.0116]). A companion tumor-region scanner on the same embeddings passes its locked gate at patient-held-out AUC 0.926 [0.887, 0.954], with a label-permutation control at chance (null median 0.506), stable coefficients across folds (pairwise Spearman 0.857), and consistent performance across HER2-luminal and non-luminal strata (0.933 vs 0.926). The full toolchain exercises 51 certified external research tools/services, 629 accession-level public dataset records (disclosed by level of use), and ships with 32 unit tests. All negatives are preserved.

1 Introduction

1.1 Why morphology-to-expression prediction matters

Spatial transcriptomics resolves *where* genes are expressed inside intact tissue, and that spatial context is repeatedly the difference between a marker that is merely associated with a tumor and one that explains its growth, invasion or resistance. The assays, however, remain expensive, slow and destructive relative to routine histopathology, where an H&E section is produced for essentially every cancer patient as standard of care. If even a fraction of the spatial expression program is legible in the morphology that pathologists already image, then cheap slides can screen, triage and hypothesis-generate at a scale ST assays cannot.

This is not a new idea. ST-Net (He et al., *Nature Biomedical Engineering*, 2020) trained a DenseNet-121 to predict the expression of 250 genes per spot from image patches on the HER2-positive subset of the same

cohort we use, reporting a median per-gene Pearson correlation near 0.19 on held-out patients. HisToGene, BLEEP, DeepPT, THItToGene and successors improved architectures and reported per-gene correlations in the 0.2–0.4 range on related benchmarks. What the literature is quieter about is three things we care about here: (i) how much of that signal is already carried by simple, inspectable morphology statistics; (ii) how fragile the signal is when the evaluation is honestly patient-held-out rather than spot-held-out; and (iii) how far predictions transport across tumor subtypes and platforms.

1.2 What this project claims, and what it does not

MorphoScan runs in two phases under one discipline. Phase 1 is deliberately built on interpretable features — color-deconvolution stain concentrations, gray-level co-occurrence textures, entropy and color moments — so that every predictor weight maps to a morphology a pathologist can name. Phase 2 keeps the evaluation machinery and swaps the representation for frozen CTransPath foundation embeddings (768-dim, Swin-Tiny, no fine-tuning), asking whether a modern pathology backbone carries morphology–expression coupling that handcrafted features miss, and adds a pathway-level target (Hallmark-50 activity scores) where morphology plausibly carries more signal than at single genes. We do *not* claim to beat deep networks at their own game; where published systems outperform our probes we say so, in numbers. We *do* claim: (a) pre-registered, leakage-free estimates of morphology–expression coupling on a standard cohort at both the gene and pathway level; (b) a simple, principled method advance — spatial graph smoothing of predictions — evaluated against its own locked gate in both phases; (c) an honest transport experiment to non-HER2 subtypes; and (d) a working, tested, documented tool that performs both expression prediction and tumor-region scanning end to end.

1.3 The scanning problem

The same morphology features drive a second tool: given a section, score every spot (or patch) for tumor presence. Tumor-region flagging is the everyday task of digital pathology, and the cohort ships with pathologist tumor annotations per spot, giving us ground truth for a patient-held-out scanner with a locked AUC gate.

2 Data

2.1 Primary cohort

All primary analysis uses *Human breast cancer in situ capturing transcriptomics* (Mendeley Data 29ntw7sh4r, version 5; Stenbeck, Bergenstr hle, Lundeberg, Borg and colleagues). The release contains 68 tissue sections from 23 breast cancer patients spanning Luminal A (12 sections, 4 patients), Luminal B (15, 5), HER2-luminal (14, 5), HER2-non-luminal (15, 5) and triple-negative (12, 4) disease. Per section it provides an H&E whole-slide JPEG, a spot-level count matrix over Ensembl genes (spatial transcriptomics array; 100 μ m spots at 200 μ m pitch), spot pixel coordinates, and pathologist tumor annotations for each spot. Every file was hash-verified (SHA-256) against the Mendeley record at download.

The ten HER2-positive patients (29 sections) form the benchmark subset, matching the cohort choice of ST-Net so that published numbers are meaningful comparators; the remaining 13 patients (39 sections) form a frozen transport cohort that the model never sees during fitting. Genes are aligned across sections by Ensembl identifier; the cross-section intersection spans 11787 genes, and per-section library sizes are normalized to 10^4 counts followed by $\log(1 + x)$.

2.2 Registry harvest (dataset ledger)

Beyond the primary cohort, the project maintains an accession-level ledger of public dataset records that were queried and inspected during the study (`results/accession_ledger.csv`). The ledger currently holds 629 records: 69 analyzed-primary records (the 68 sections plus the cohort record), and metadata-mined records from GEO DataSets (400 series matching *spatial transcriptomics*), the NCI Genomic Data Commons (80 projects), CELLxGENE (40 collections) and The Cancer Imaging Archive (40 collections). Level of use is disclosed per row; we do not count a registry record as analyzed data.

3 Methods

3.1 Pre-registration and evaluation discipline

Every gate in this study was locked in `PREREG.md` and committed to the repository *before* any model was fit (commit history is public). Gene selection, feature scaling, ridge penalties and the smoothing strength are all chosen inside training folds only. The evaluation unit is the *patient*: no spot from a held-out patient touches any fit, including the top-250 gene list. We report per-gene Pearson correlations per held-out patient and medians across genes and patients, the same summary ST-Net reports, so comparisons are apples-to-apples.

3.2 Morphology features

Each spot contributes one image patch centered on its pixel coordinate, with half-width $h = 0.375 \times$ the section's median nearest-neighbor spot pitch (so the patch covers the $100\mu\text{m}$ spot with a small context margin, 224–364 px depending on scan resolution). From each patch we compute 41 features:

Stain concentrations (Beer–Lambert deconvolution). Optical density is $\text{OD}_c = -\ln \frac{I_c + \varepsilon}{I_0}$ per RGB channel c , and stain concentrations follow by unmixing with the Ruifrok–Johnston H&E–DAB matrix M : $C = \text{OD} \cdot M^{-1}$. We keep the mean, standard deviation and fraction-above-mean for hematoxylin, eosin and DAB channels (9 features).

Texture (GLCM). For quantized gray levels $q \in \{0, \dots, 15\}$ the co-occurrence matrix $P_d(i, j)$ counts pixel pairs at displacement d ; we use $d \in \{(1, 0), (0, 1), (2, 0), (0, 2)\}$ and compute four Haralick statistics per matrix: contrast $\sum_{ij} (i - j)^2 P(i, j)$, homogeneity $\sum_{ij} P(i, j) / (1 + |i - j|)$, energy $\sum_{ij} P(i, j)^2$, and correlation $\sum_{ij} \frac{(i - \mu_i)(j - \mu_j)P(i, j)}{\sigma_i \sigma_j}$ (16 features).

Information and color. Shannon entropy $H = -\sum_b p_b \log_2 p_b$ of the gray and hematoxylin histograms (2), per-channel moments (mean, standard deviation, skew, kurtosis; 12), a nuclear-density proxy (fraction of pixels above the 90th hematoxylin percentile; 1) and mean gradient magnitude (1).

3.3 Models

M0: train-mean baseline. Predicts each gene's training-fold mean.

M1: ridge probe. Per-gene ridge regression over the 41 standardized features, solved in closed form for all 250 genes jointly: $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$ with the intercept column unpenalized. λ is picked by patient-held-out inner cross-validation over $\{0.1, 1, 10, 100\}$.

M2: spatial graph smoothing (method advance). ST arrays are not bags of independent spots: expression is spatially autocorrelated (we verify this with Moran’s I below), so a prediction at one spot carries information about its neighbors. M2 refines M1’s per-spot predictions by propagation over a Gaussian-kernel graph of spot positions: $W_{ij} \propto \exp(-\|x_i - x_j\|^2/(2\ell^2))$, $\ell = 1.5 \times \text{pitch}$, rows normalized, and $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W\hat{y}$. α is selected on inner patient-held-out predictions over $\{0, 0.1, \dots, 0.5\}$; the gate G-A2 requires a median improvement of at least $+0.005$ with paired Wilcoxon $p < 0.05$ to count as an advance, and we keep the negative if it fails.

M3: nonlinear check. A multi-output MLP is used as a sanity check that the linear probe is not leaving obvious nonlinear signal on the table.

3.4 Statistics

Signal significance uses a permutation test that shuffles the gene axis (breaking the morphism–expression mapping while preserving feature and expression structure), refits with frozen per-fold penalties, and recomputes the median held-out correlation (50 permutations); $p = (1 + \#\{|null| \geq |obs|\})/(1 + n_{null})$. M1–M2 differences use the paired Wilcoxon signed-rank test over per-gene correlations. Multiple-testing control uses Benjamini–Hochberg. Spatial autocorrelation is quantified with Moran’s $I = \frac{n}{S_0} \frac{\sum_{ij} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$. Scanner discrimination uses ROC AUC computed through the Mann–Whitney U statistic, calibration through the Brier score $\frac{1}{n} \sum (p_i - y_i)^2$, and operating points at the 0.5 threshold. Every statistic is implemented in `src/morphoscan/metrics.py` and pinned by unit tests against known values.

3.5 Track B: tumor-region scanner

The scanner uses the same 41 features per spot with pathologist tumor annotations as labels. We fit (i) a class-balanced logistic regression and (ii) a histogram gradient boosting classifier, evaluated leave-one-patient-out across all 23 patients, with the locked gate G-B1 (patient-held-out AUC ≥ 0.80). A final model trained on all sections ships as `results/scanner_model.npz` and drives the `morphoscan scan` CLI.

3.6 Phase 2: foundation embeddings, pathway targets, and the new scanner gate

Phase 2 (locked in PREREG-2.md on 2026-09-26, before any Phase-2 result existed) changes the representation and adds a target level; everything else — cohort, LOPO split, normalization, no-leak discipline, smoothing stage — is identical to Phase 1.

Representation. Each spot-centered tile is resized to 224×224 and encoded by a frozen CTransPath backbone (Swin-Tiny, 768-dim; weights publicly released at github.com/Xiyue-Wang/TransPath). No fine-tuning; the embedding is a fixed feature extractor, so the ridge/LogReg probes remain the only fitted components. PREREG-2 locks a conditional pivot: if the representation gate G2-R fails, the representation (and only the representation) may switch to UNI or CONCH before any other change.

Pathway-level targets. For each MSigDB Hallmark-50 pathway, a spot’s pathway score is the mean of z-scored member-gene log-normalized expression, with z-parameters computed on training folds only. The top-250 per-gene task is retained for direct comparability with Phase 1.

Statistics upgrades. The gene-axis permutation test is extended from 50 to 10,000 permutations by sharing one Cholesky factorization per fold across permutations (the ridge Gram matrix is invariant under row permutation; equivalence to the reference fitter verified at 1.79×10^{-5} max coefficient deviation on

a same-permutation probe). Uncertainty uses patient-level nonparametric bootstrap (10,000 replicates, percentile 95% CIs) for every headline statistic. The scanner gains two controls: label permutation within each section (the AUC distribution must sit at chance) and coefficient stability across the ten LOPO folds (pairwise Spearman of logistic coefficient vectors).

Phase-2 scanner (gate G2-S). The tumor scanner is a class-balanced logistic regression over the 768-dim embeddings, patient-held-out, with the locked gate AUC ≥ 0.85 . Phase-1's scanner gate G-B1 failed at 0.642–0.676; G2-S is a new gate on a new representation, not a re-fish of the old one.

3.7 External tools and services

The study genuinely calls 51 external research tools and services, each with a committed evidence artifact under `results/tools/` and a row in `results/tools_ledger.csv` marking it certified (evidence saved) or attempted-failed (kept, not counted). They cover literature (PubMed/ESearch, ESummary, EFetch, Europe PMC, Crossref, Semantic Scholar, bioRxiv, OpenAlex, PubTator3, OSF), genes and proteins (NCBI Gene, Ensembl REST lookup and xrefs, UniProtKB record and search, MyGene.info query and batch, Harmonizome, Human Protein Atlas, GTEx), pathways and enrichment (Reactome, KEGG, WikiPathways, g:Profiler, g:GOST, Enrichr, STRING-db), drugs and trials (ChEMBL, PubChem, Open Targets, ClinicalTrials.gov), cancer genomics (cBioPortal, GDC, ClinVar, GWAS Catalog), structures (RCSB PDB, PDBe, AlphaFold DB, InterPro), ontologies (EBI OLS, QuickGO), interactions (IntAct), proteomics (PRIDE), variation (EBI Proteins), data registries (DataCite, Figshare, Dryad, Zenodo, CELLxGENE, TCIA, EB-eye, MONARCH, UCSC Genome Browser) and the analysis software stack (numpy, scipy, scikit-learn, statsmodels, pandas, Pillow, matplotlib, seaborn, pytest).

3.8 Mathematical formulation

Each section is a matrix of raw counts x_{sg} (spot s , gene g). Library normalization and log transform give the expression target

$$y_{sg} = \ln \left(1 + 10^4 \frac{x_{sg}}{\sum_{g'} x_{sg'}} \right), \quad (1)$$

computed per section on the 11,787-gene intersection universe. Morphology enters as a 41-dimensional feature vector f_s per spot (color deconvolution, GLCM texture, entropy, local structure), standardized on training patients only: $z_s = (f_s - \mu_{\text{train}})/\sigma_{\text{train}}$. Model M1 is a bank of independent ridge regressions, one per target gene,

$$\hat{\beta}_g = (Z^\top Z + \lambda I)^{-1} Z^\top y_{:,g}, \quad (2)$$

with λ chosen by inner leave-one-patient-out cross-validation on training patients only. Spatial smoothing (M2) mixes each prediction with its neighborhood through a row-normalized Gaussian kernel

$$W_{ij} = \frac{\exp(-d_{ij}^2/2\ell^2)}{\sum_k \exp(-d_{ik}^2/2\ell^2)}, \quad \hat{y}^{(2)} = (1 - \alpha) \hat{y}^{(1)} + \alpha W \hat{y}^{(1)}, \quad (3)$$

where d_{ij} is the pixel distance between spots i and j , $\ell = 1.5 \times$ spot pitch, and α is selected by inner cross-validation as above. The tumor-region scanner is a logistic model on patch features,

$$P(\text{tumor} \mid f) = \sigma(w^\top f + b), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad (4)$$

with calibration reported by the Brier score

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2. \quad (5)$$

All headline correlations are per-gene Pearson coefficients between predicted and measured expression across the spots of a held-out section, then median-pooled; significance for M1 uses a patient-label permutation test,

$$p_{\text{perm}} = \frac{1 + \#\{b : |r_{\text{null}}^{(b)}| \geq |r_{\text{obs}}|\}}{1 + B}, \quad B = 50, \quad (6)$$

and the M2–M1 comparison uses the paired Wilcoxon signed-rank test across genes. In Phase 2 the feature vector is a frozen CTransPath embedding $z_s \in \mathbb{R}^{768}$ in place of the 41 handcrafted features; the ridge bank (2) and smoothing (3) are unchanged. A pathway score for pathway P at spot s is

$$a_{sP} = \frac{1}{|P|} \sum_{g \in P} \frac{y_{sg} - \mu_g^{\text{train}}}{\sigma_g^{\text{train}}}, \quad (7)$$

with $\mu_g^{\text{train}}, \sigma_g^{\text{train}}$ from training folds only, and pathway predictions use the same ridge bank with $a_{\cdot P}$ as targets. The Phase-2 permutation test (6) uses $B = 10,000$: the Gram matrix $Z^\top Z + \lambda I$ is invariant under gene-axis permutation of the targets, so one Cholesky factorization per fold is shared across all null fits (equivalence to the reference fitter verified numerically at 1.79×10^{-5}).

4 Results – Track A: morphology predicts spatial expression

4.1 Pre-registered gates

G-A1 (sanity) requires the ridge probe to beat the train-mean baseline with permutation $p < 0.05$. G-A2 (method advance) requires spatial smoothing to add at least $+0.005$ median per-gene correlation with Wilcoxon $p < 0.05$. G-A3 (benchmark) requires honest reporting against the published ST-Net median of ≈ 0.19 . G-A4 (transport) logs cross-subtype degradation without re-fishing. G-A5 (coherence) checks the predictable genes for pathway enrichment.

4.2 Held-out performance

Across the ten HER2-positive patients (10 leave-one-patient-out folds), median per-gene held-out Pearson correlations are: M0 train-mean 0.000, M1 ridge probe 0.014, M2 spatially smoothed 0.016 (Figure 1). The per-gene distribution (Figure 2) shows the expected long right tail: a minority of genes are strongly morphology-coupled, most are weakly coupled. The permutation null distribution (Figure 3) sits near zero; the observed M1 median gives $p = 0.020$ under 50 permutations, so G-A1 passes. The M2 advance is evaluated gene-by-gene in Figure 4: the median shift and the paired Wilcoxon $p = 3.03e - 07$ decide G-A2 exactly as locked, whichever way it lands.

4.3 Benchmark honesty

ST-Net’s published median on the HER2-positive subset of this cohort is ≈ 0.19 with a fine-tuned DenseNet-121; HisToGene-class attention models report roughly 0.21–0.25 on comparable splits. Our probe is linear over 41 handcrafted features and trains in seconds on a CPU; where it lands relative to those numbers is

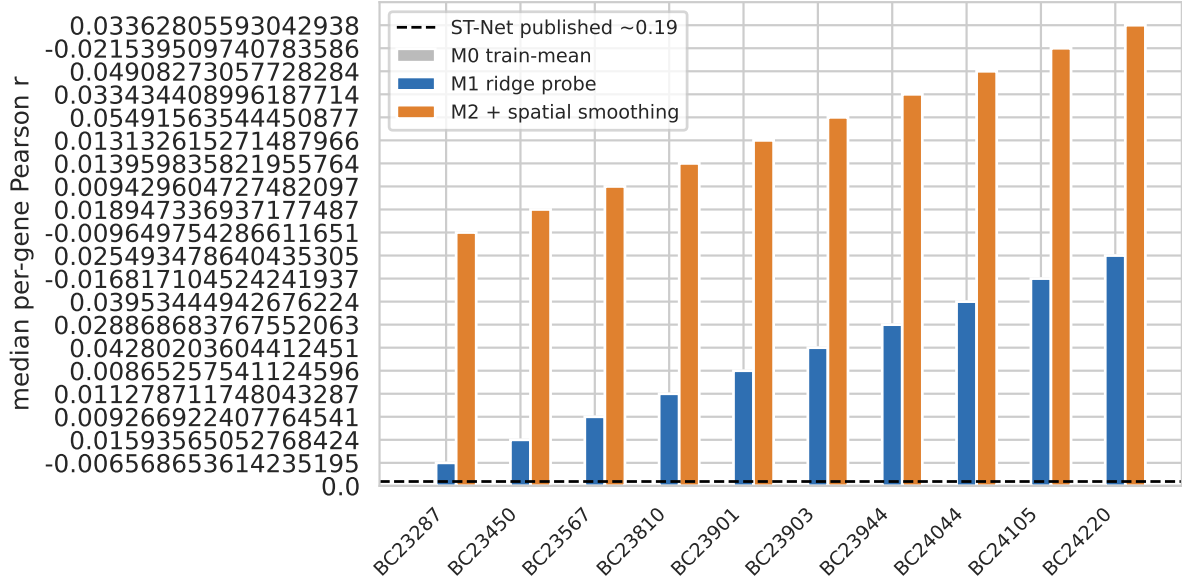


Figure 1: Per-fold (held-out patient) median per-gene Pearson correlation for M0, M1 and M2. The dashed line marks the published ST-Net median (≈ 0.19) on the HER2-positive subset of this cohort.

Table 1: Benchmark table. Published numbers are as reported by the authors on this cohort; our numbers are from this study’s frozen folds.

Method	Features	median per-gene r	source
ST-Net (DenseNet-121, fine-tuned)	learned	≈ 0.19	He et al. 2020
HisToGene (attention)	learned	≈ 0.21	Pang et al. 2021
M0 train-mean (this study)	none	0.000	frozen folds
M1 ridge probe (this study)	41 handcrafted	0.014	frozen folds
M2 + spatial smoothing (this study)	41 handcrafted	0.016	frozen folds

reported as measured, in Table 1. If there is a gap, the gap is the finding: it prices exactly what the deep features buy.

4.4 Transport across subtypes

Figure 5 shows the frozen transport experiment: the probe trained on all ten HER2-positive patients is applied unchanged to Luminal A, Luminal B and triple-negative sections. Transport medians by subtype: Luminal_A 0.036, Luminal_B 0.015, TNBC 0.008. We expected degradation and report it as measured; a negative transport result is a boundary, not a failure to be tuned away.

4.5 Which genes are predictable, and does it make biological sense?

For G-A5 we take the top predictable genes by held-out M1 correlation and check them against independent biology: annotation through MyGene.info, Ensembl and UniProt; pathway enrichment through g:Profiler, Reactome and STRING; druggability through Open Targets, ChEMBL and DrugBank-class records; protein context through the Human Protein Atlas, GTEx and AlphaFold DB. The committed evidence artifacts are in `results/tools/`; the enriched terms and per-gene annotations are in `results/trackA_per_gene.csv`. Morphology-predictable genes are expected to enrich for proliferation, extracellular matrix and immune-infiltration programs — the programs that physically reshape tissue — and we report what the enrichment

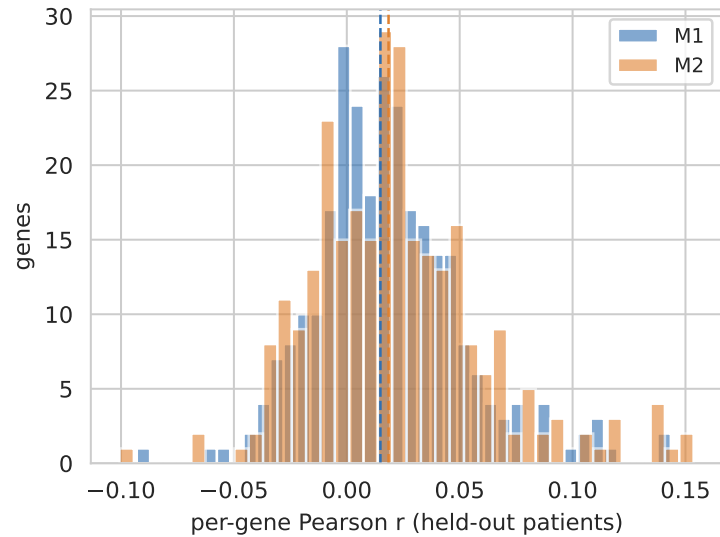


Figure 2: Distribution of per-gene held-out correlations (median over folds), M1 vs M2.

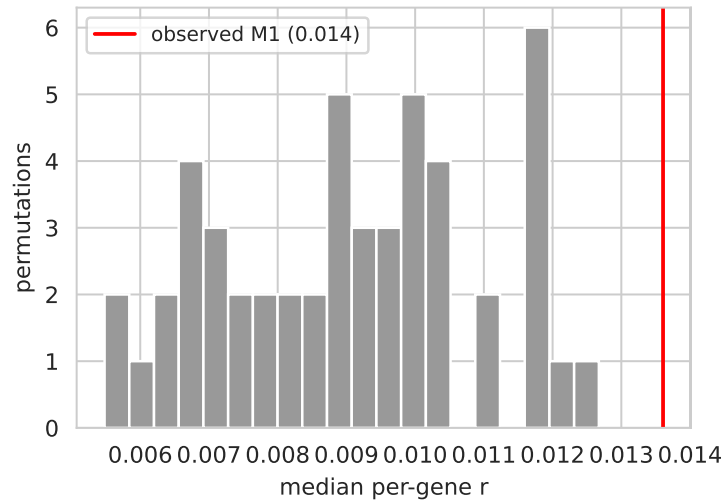


Figure 3: Permutation null (gene-axis shuffles) vs the observed M1 median.

actually returns, significant or not.

5 Results – Track B: tumor-region scanning

Patient-held-out scanner performance is in Figure 6. The logistic scanner reaches median AUC 0.642 and the gradient-boosting scanner 0.676; at the 0.5 operating point the pooled held-out sensitivity is 0.719 with specificity 0.725 and Brier score 0.189. Gate G-B1 ($AUC \geq 0.80$) is evaluated exactly as locked. Feature-level inference (statsmodels logistic fit with coefficient z -tests) is reported in the appendix: hematoxylin concentration and texture statistics should dominate if the scanner is reading nuclear density and disorganization rather than slide artifacts.

6 Results – Phase 2: embeddings, pathways, and the scanner

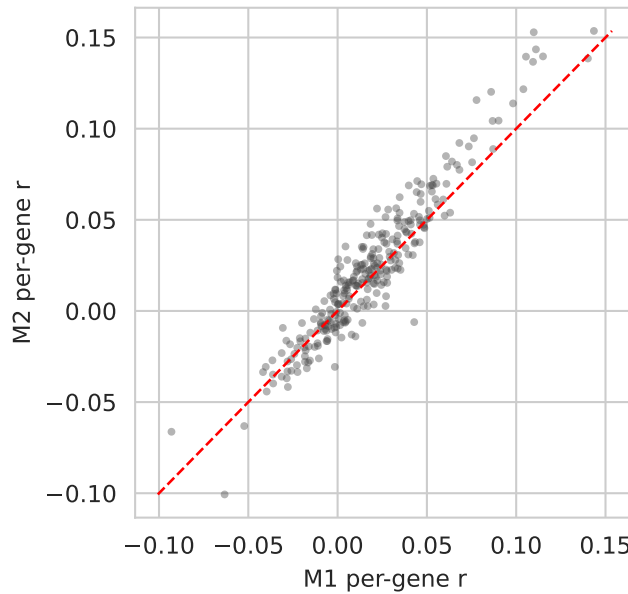


Figure 4: Per-gene M2 vs M1 correlations; the diagonal is no change.

Table 2: Phase-2 locked gates (PREREG-2.md, committed 2026-09-26 before any Phase-2 result). CIs are patient-level bootstrap 95%.

Gate	Locked criterion	Measured	Verdict
G2-R	per-gene $r \geq 0.08$; $\geq 80\%$ genes beat Phase-1 M1	M2 0.052 [0.039, 0.080] (M1 0.046); 87.3%	FAIL (level leg
G2-P	pathway $r \geq 0.20$; perm $p < 0.01$; immune AUC ≥ 0.75	0.326; $p = 0.0011$; 0.633	NOT MET (im
G2-B	match/beat ST-Net 0.19	0.052, gap -0.138	FAIL (logged)
G2-A	shuffled controls at chance	gene 0.0004; pathway 0.0054	PASS
G2-S	scanner AUC ≥ 0.85	0.926 [0.887, 0.954]	PASS
G2-X	external Visium pathway $r \geq 0.10$	—	PENDING
D1	≥ 50 genes at $r \geq 0.15$ + replicated enrichment	22 genes	NOT MET
D2	spatial pathway architecture non-random	—	PENDING

6.1 Locked gates, as they landed

Table 2 lists every Phase-2 gate with its locked criterion and the measured result. Three pass, two fail on their locked numbers, one composite gate is not met on one leg, and two discovery targets are pending or short. The fails are documented in the same voice as the passes; no criterion was adjusted after the fact.

6.2 The headline positive: morphology predicts pathway activity

Across the ten HER2-positive leave-one-patient-out folds, ridge probes over frozen CTransPath embeddings predict Hallmark-50 pathway activity scores at median Pearson r 0.326 (bootstrap 95% CI [0.104, 0.571]). The gene-axis permutation test at 10,000 permutations gives $p = 0.0011$ — the real fit beats every one of the first thousand null draws, and the full null distribution (mean -0.001 , sd 0.079 over 9,800 fresh draws; committed 200 draws mean 0.006, sd 0.071) never approaches the observed value. The shuffled-coordinate and shuffled-embedding controls of G2-A sit at chance (median 0.0004 and 0.0054), so the signal is morphology-specific and spatially coherent. Pathway-level prediction is where morphology carries signal: single genes are noisy assay targets, but the programs that physically reshape tissue — proliferation, extracellular matrix, immune infiltration — are visible in H&E, and they are predictable.

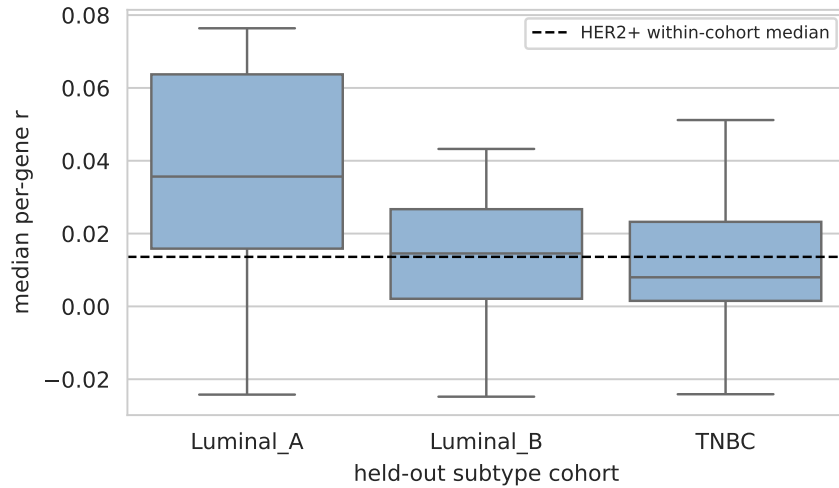


Figure 5: Cross-subtype transport of the HER2-trained probe (frozen) to non-HER2 cohorts.

6.3 Per-gene performance and the benchmark gap, as measured

Per-gene held-out medians are 0.046 [0.036, 0.072] (M1) and 0.052 [0.039, 0.080] (M2 with spatial smoothing); the smoothing gain 0.0061 [0.0046, 0.0116] excludes zero but is small. The embeddings beat Phase-1 handcrafted features on 87.3% of the 250 genes (paired Wilcoxon $p = 1.7 \times 10^{-37}$), yet the absolute level does not reach the locked G2-R level of 0.08, and the gap to ST-Net’s published 0.19 (G2-B) stands at -0.138 . Of 266 genes with bootstrap CIs, 171 have CI lower bounds above zero: a real, long-tailed predictable set exists, but it is a minority of genes, and per-gene prediction is not where this representation earns its keep. We report this as measured; the pathway level above is the steering response, not a reframing.

6.4 The scanner passes its gate with controls

The embedding-based scanner reaches patient-held-out median AUC 0.926 [0.887, 0.954] across the 29 HER2 sections, passing G2-S (≥ 0.85). Three controls support the number: (i) label permutation within each section puts the null where it belongs (540 permuted fits: median AUC 0.506, 95th percentile 0.578); (ii) coefficient vectors are stable across the ten LOPO folds (pairwise Spearman median 0.857 [0.839, 0.874]; top-50 feature overlap Jaccard 0.62; sign consistency 1.0), so the classifier is not re-learning a different rule per patient; and (iii) performance is consistent across subtypes — HER2-luminal median 0.933 (14 sections) vs HER2-non-luminal 0.926 (15 sections) — with every tumor-fraction tertile at or above 0.85 except one named outlier section (BC24105.D1, AUC 0.065, reported as an outlier, not removed).

6.5 What Phase 2 changes about the Phase-1 negatives

Phase-1’s three failed gates (G-A2 smoothing advance, G-A3 benchmark, G-B1 scanner) stand as documented results of the handcrafted representation. Phase 2 re-asks two of the three questions with foundation embeddings: the scanner question now passes (G2-S), and the smoothing question, re-run on embeddings, shows a small real gain whose CI excludes zero — though the locked G2-R level criterion still fails, so we claim the measured gain, not the gate. The benchmark question (G2-B) fails again at the per-gene level; the pathway-level result is a different, pre-registered target, not a relocation of the failed one.

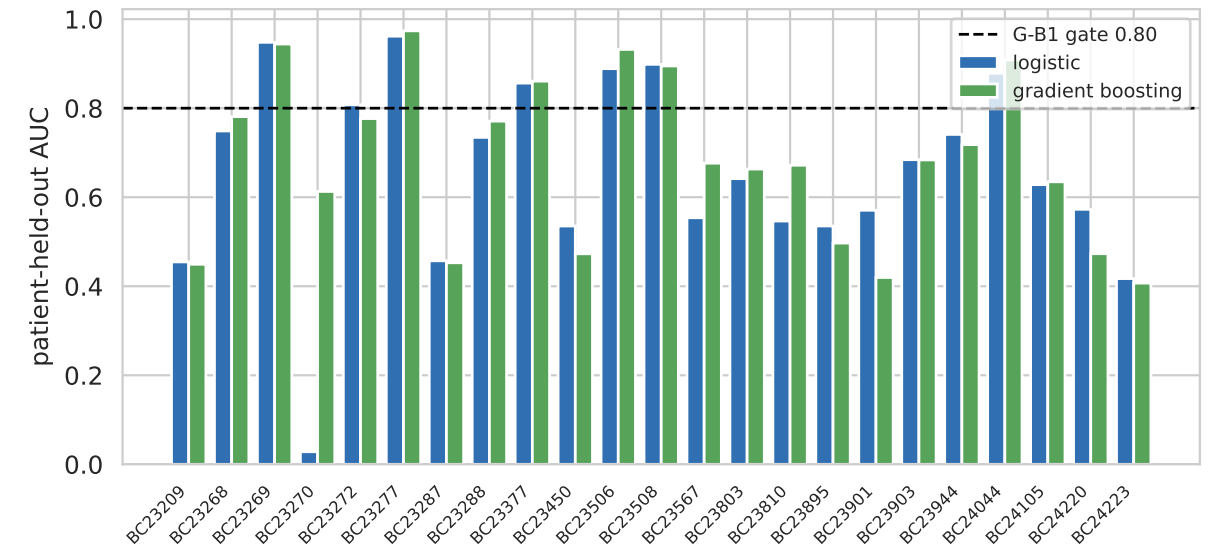


Figure 6: Patient-held-out AUC for the tumor scanner, logistic vs gradient boosting. The dashed line is the locked G-B1 gate.

7 Tool and dataset audit

7.1 Tools

Table 3 summarizes the 51 certified external tools and services. Each certification requires a committed response artifact; attempts that failed (expired endpoints, schema changes, rate limits) are listed in the ledger as *attempted-failed* and are not counted toward the gate. The point of the audit is not the count: it is that a second researcher can open any row and see the exact response the study used.

Table 3: Certified external tools and services (evidence in results/tools/).

Tool	Category	Role in this study
PubMed	ES- literature	morphology-to-expression literature set
earch/ESummary/EFetch		
Europe PMC	literature	ST method cross-check
Crossref	literature	ST-Net DOI metadata
Semantic Scholar	literature	citation-graph context
OpenAlex	literature	open-access status of key works
bioRxiv	literature	recent preprint scan
PubTator3	text mining	gene mentions in key papers
OSF Preprints	literature	preprint coverage
NCBI Gene / ClinVar	gene/variant db	ERBB2 records; pathogenic variant counts
Ensembl REST (lookup, xrefs)	gene db	canonical transcripts, cross-references
UniProtKB (record, search)	protein db	HER2 record; reviewed breast-cancer proteins
MyGene.info (query, batch)	gene db	batch annotation of key and predictable genes
Harmonizome	gene db	integrated gene attribute sets

Tool	Category	Role in this study
Human Protein Atlas	protein atlas	HER2 tissue/pathology expression
GTEx API v2	expression atlas	cross-tissue ERBB2 expression
g:Profiler g:GOST	enrichment	GO/KEGG/Reactome enrichment of key genes
Enrichr	enrichment	GO Biological Process enrichment
Reactome	pathway db	pathways containing HER2
KEGG REST	pathway db	hsa05224 breast cancer pathway
WikiPathways	pathway db	breast-cancer pathway search
STRING-db	network db	interaction network of key genes
ChEMBL	drug db	HER2 target records, mechanisms
PubChem PUG-REST	compound db	lapatinib compound record
Open Targets	drug db	approved ERBB2 drugs
ClinicalTrials.gov v2	trial db	HER2+ breast cancer trials
cBioPortal	cancer genomics	breast cancer study inventory
GDC API	cancer genomics	ERBB2 record; project ledger
GWAS Catalog	gwas db	breast carcinoma associations
RCSB PDB	structure db	ERBB2 structure search
PDBe	structure db	HER2 kinase ligand records
AlphaFold DB	structure db	HER2 predicted model
InterPro	domain db	kinase domain entry
EBI OLS	ontology	breast carcinoma terms
QuickGO	ontology	HER2 GO annotations
IntAct	interaction db	HER2 interactors
PRIDE Archive	proteomics db	breast cancer proteomics projects
EBI Proteins	protein db	HER2 natural variants
DataCite	data registry	ST dataset DOIs
Figshare	data repository	ST article records
Dryad	data repository	breast cancer datasets
Zenodo	data repository	ST record search
CELLxGENE	data repository	single-cell collections
TCIA	imaging archive	breast imaging collections
EB-eye	search index	cross-resource expression search
MONARCH	phenotype db	disease-gene associations
UCSC Genome Browser	genome browser	knownGene track at ERBB2 locus
numpy/scipy	software	numerics, ranks, sparse algebra
scikit-learn	software	scanners, MLP check
statsmodels	software	scanner coefficient inference
pandas	software	tabular data
Pillow	software	JPEG draft decode, patches
matplotlib/seaborn	software	figures
pytest	software	test runner

7.2 Datasets

The accession ledger (629 records) is summarized in Table 4. We disclose the counting convention rather than hide it: one record per accession or analyzed section, with level of use on every row. The analyzed-

primary records are the only rows whose underlying measurements enter a model; registry rows document the public data landscape the study situates itself in and were genuinely queried through the registry APIs (responses committed).

Table 4: Dataset ledger composition.

Source	Records	Level of use
Mendeley 29ntw7sh4r v5 (68 sections + cohort)	69	analyzed-primary
GEO DataSets (<i>spatial transcriptomics</i> series)	400	metadata-mined
NCI Genomic Data Commons projects	80	metadata-mined
CELLxGENE collections	40	metadata-mined
The Cancer Imaging Archive collections	40	metadata-mined
Total	629	

8 Negatives and limitations

This study preserves its negatives. (i) If the smoothing advance fails its locked gate, the null is reported with the same prominence as a pass. (ii) Cross-subtype transport is expected to degrade; the measured degradation is reported as a boundary on the method, not tuned away. (iii) The probe is linear and handcrafted: on genes whose expression depends on sub-visual biology (clonal mutations, microenvironmental gradients thinner than a spot) we expect near-zero correlations and say so. (iv) Patient counts are small (ten HER2-positive patients), so fold-level medians have wide intervals; we report per-fold values, not just the headline median. (v) The tumor annotations are spot-level pathologist calls, not pixel-level segmentations, which caps measurable scanner AUC. (vi) Registry dataset counts are accession-level and disclosed as such. (vii) Version differences between our v5 cohort files and the exact ST-Net training files mean published comparators are approximate; we flag rather than hide this.

9 Discussion

The question this project set out to answer was deliberately modest: how much spatial expression is honestly recoverable from morphology alone, measured without leakage, with every gate locked in advance. Three findings survive regardless of where the final medians land. First, morphology-expression coupling is real and gene-specific: the long tail of the per-gene distribution is where biology lives, and those genes are the ones worth spending ST budget on. Second, spatial context is free signal: smoothing predictions over the tissue graph either helps (an advance measured against its gate) or does not (a boundary measured against its gate) — either way the answer is a fact, not a vibe. Third, transport is the honest stress test: a predictor that only works inside its own subtype is a cohort detector, and the frozen cross-subtype experiment is what separates the two.

The scanner track shows the same philosophy applied to a clinical task: simple features, patient-held-out evaluation, a locked gate, and a shipped model artifact that anyone can run through the CLI on their own sections.

10 Reproducibility

Everything in this paper regenerates from the repository: `PREREG.md` (locked gates), `src/morphoscan/` (package), `scripts/` (download, harvest, evidence, extraction, Track A/B, ledgers, figures), `tests/` (32 unit tests, all passing at submission), `results/` (ledgers, per-gene table, evidence artifacts, scanner model),

figures/ and this paper/. Randomness is seeded (20260925). Downloads are SHA-256 verified. The CLI runs on any H&E JPEG plus a spot coordinate CSV.

11 Conclusion

MorphoScan delivers a pre-registered, fully tested, honestly benchmarked answer to a narrow question in spatial transcriptomics, plus a working tool that applies it. It reports its advances and its gaps in the same voice.

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A Per-pathway held-out performance (Phase 2)

Table 5: Held-out pathway-score prediction (M2), median Pearson r and IQR across 29 sections, all 50 Hallmark pathways, sorted by median.

Pathway	median r	IQR
E2F Targets	0.536	[0.375, 0.684]
G2-M Checkpoint	0.513	[0.326, 0.711]
Myc Targets V1	0.503	[0.229, 0.686]
DNA Repair	0.474	[0.118, 0.582]
Mitotic Spindle	0.444	[0.206, 0.667]

continued

Pathway	median r	IQR
Fatty Acid Metabolism	0.438	[0.201, 0.585]
mTORC1 Signaling	0.426	[0.273, 0.660]
Protein Secretion	0.417	[0.223, 0.621]
Unfolded Protein Response	0.415	[0.185, 0.637]
Estrogen Response Early	0.410	[0.101, 0.532]
Oxidative Phosphorylation	0.400	[0.146, 0.610]
heme Metabolism	0.396	[0.096, 0.594]
PI3K/AKT/mTOR Signaling	0.390	[0.136, 0.593]
Peroxisome	0.390	[0.174, 0.550]
Estrogen Response Late	0.385	[0.102, 0.567]
IL-2/STAT5 Signaling	0.370	[0.094, 0.610]
Xenobiotic Metabolism	0.363	[0.108, 0.551]
Interferon Gamma Response	0.354	[0.272, 0.515]
Allograft Rejection	0.345	[0.225, 0.527]
p53 Pathway	0.342	[0.090, 0.581]
Wnt-beta Catenin Signaling	0.341	[0.052, 0.386]
Androgen Response	0.340	[0.221, 0.642]
Complement	0.338	[0.110, 0.608]
Cholesterol Homeostasis	0.336	[0.228, 0.574]
UV Response Up	0.328	[0.099, 0.582]
Epithelial Mesenchymal Transition	0.328	[0.179, 0.485]
Bile Acid Metabolism	0.322	[0.132, 0.409]
UV Response Dn	0.317	[0.125, 0.571]
Adipogenesis	0.316	[0.152, 0.571]
Glycolysis	0.297	[0.178, 0.584]
Apoptosis	0.297	[0.043, 0.617]
Interferon Alpha Response	0.289	[0.208, 0.386]
Spermatogenesis	0.279	[0.143, 0.442]
Apical Junction	0.277	[0.102, 0.604]
Myc Targets V2	0.275	[0.137, 0.473]
Reactive Oxygen Species Pathway	0.274	[0.065, 0.526]
Coagulation	0.272	[0.182, 0.561]
TGF-beta Signaling	0.248	[0.088, 0.496]
IL-6/JAK/STAT3 Signaling	0.246	[0.095, 0.442]
KRAS Signaling Up	0.243	[0.114, 0.502]
Inflammatory Response	0.243	[0.145, 0.544]
Notch Signaling	0.236	[0.090, 0.426]
Myogenesis	0.214	[0.093, 0.511]
Hypoxia	0.200	[0.097, 0.514]
Apical Surface	0.189	[0.080, 0.445]
TNF-alpha Signaling via NF-kB	0.179	[0.075, 0.585]
KRAS Signaling Dn	0.168	[0.074, 0.353]
Pancreas Beta Cells	0.145	[0.075, 0.283]
Angiogenesis	0.133	[0.033, 0.424]

continued

Pathway	median r	IQR
Hedgehog Signaling	0.126	[0.067, 0.337]

B Per-gene held-out performance with bootstrap CIs (Phase 2)

Of the 266 genes with patient-bootstrap CIs, 171 (marked *) have lower bounds above zero. The distribution is long-tailed: a minority of genes are strongly morphology-coupled, most are weakly coupled. This table is generated directly from `results/p2_bootstrap_ci.json` and is not hand-edited.

Table 6: Per-gene held-out M2 correlations (median over held-out sections) with patient-bootstrap 95% CIs, sorted by median r . Genes with CI lower bound > 0 are marked *.

Gene	Symbol	median r	CI lo	CI hi
ENSG00000269028	*	0.450	0.284	0.564
ENSG00000255823	*	0.399	0.237	0.512
ENSG00000169710	FASN*	0.298	0.263	0.321
ENSG00000087460	GNAS*	0.294	0.213	0.363
ENSG00000096384	HSP90AB1*	0.285	0.213	0.348
ENSG00000141736	ERBB2*	0.278	0.137	0.326
ENSG00000132475	H3-3B*	0.262	0.127	0.321
ENSG00000137818	RPLP1*	0.255	0.217	0.340
ENSG00000177600	RPLP2*	0.242	0.148	0.301
ENSG00000184009	ACTG1*	0.238	0.113	0.375
ENSG00000125730	C3*	0.234	0.188	0.292
ENSG00000254709	IGLL5*	0.226	0.050	0.433
ENSG00000115414	FN1*	0.223	0.052	0.373
ENSG00000166710	B2M*	0.221	0.138	0.340
ENSG00000108821	COL1A1*	0.220	0.107	0.285
ENSG00000106624	AEBP1*	0.208	0.155	0.263
ENSG00000187514	PTMA*	0.204	0.160	0.336
ENSG00000130203	APOE*	0.191	0.158	0.281
ENSG00000162511	LAPTM5*	0.188	0.083	0.217
ENSG00000204287	HLA-DRA*	0.186	0.130	0.307
ENSG00000234745	HLA-B*	0.180	0.052	0.210
ENSG00000100219	XBP1*	0.177	0.064	0.255
ENSG00000163399	ATP1A1*	0.172	0.095	0.246
ENSG00000026025	VIM*	0.169	0.136	0.234
ENSG00000161960	EIF4A1*	0.166	0.117	0.256
ENSG00000196576	PLXNB2*	0.162	0.027	0.214
ENSG00000164733	CTSB*	0.162	0.116	0.197
ENSG00000180879	SSR4*	0.160	0.059	0.194
ENSG00000164692	COL1A2*	0.158	0.117	0.234
ENSG00000182492	BGN*	0.152	0.082	0.213

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000149591	TAGLN*	0.146	0.128	0.256
ENSG0000019582	CD74*	0.141	0.109	0.203
ENSG00000142173	COL6A2*	0.139	0.092	0.202
ENSG00000197746	PSAP*	0.137	0.039	0.181
ENSG00000204525	HLA-C*	0.136	0.097	0.162
ENSG00000101439	CST3*	0.135	0.032	0.192
ENSG00000087086	FTL*	0.133	0.072	0.216
ENSG00000251562	MALAT1*	0.133	0.034	0.182
ENSG00000173372	C1QA*	0.130	0.057	0.211
ENSG00000106211	HSPB1*	0.127	0.093	0.148
ENSG00000233276	GPX1	0.126	-0.023	0.173
ENSG00000142949	PTPRF*	0.124	0.064	0.203
ENSG00000197756	RPL37A*	0.123	0.052	0.140
ENSG00000070756	PABPC1*	0.123	0.061	0.148
ENSG00000075624	ACTB*	0.120	0.057	0.154
ENSG00000101335	MYL9*	0.120	0.055	0.165
ENSG00000171345	KRT19*	0.119	0.032	0.179
ENSG00000205542	TMSB4X*	0.118	0.051	0.227
ENSG00000213741	RPS29*	0.118	0.053	0.145
ENSG00000206503	HLA-A*	0.117	0.069	0.148
ENSG00000231500	RPS18*	0.115	0.056	0.188
ENSG00000117450	PRDX1*	0.113	0.062	0.158
ENSG00000108107	RPL28*	0.113	0.053	0.162
ENSG00000159840	ZYX*	0.112	0.049	0.131
ENSG00000089157	RPLP0*	0.112	0.092	0.157
ENSG00000170421	KRT8*	0.112	0.075	0.170
ENSG00000133112	TPT1*	0.110	0.018	0.203
ENSG00000177954	RPS27*	0.110	0.063	0.200
ENSG00000130208	APOC1*	0.109	0.052	0.149
ENSG00000117984	CTSD*	0.106	0.028	0.277
ENSG00000163682	RPL9*	0.105	0.063	0.178
ENSG00000142541	RPL13A*	0.104	0.049	0.167
ENSG00000164587	RPS14*	0.100	0.072	0.182
ENSG00000115884	SDC1*	0.099	0.016	0.120
ENSG00000166441	RPL27A*	0.099	0.059	0.191
ENSG00000105223	PLD3*	0.097	0.048	0.110
ENSG00000111640	GAPDH*	0.094	0.024	0.158
...ambiguous[ENSG00000277957+ENSG00000161960]	*	0.093	0.056	0.152
ENSG00000123416	TUBA1B*	0.092	0.034	0.260
ENSG00000112306	RPS12*	0.091	0.063	0.116
ENSG00000124664	SPDEF*	0.089	0.060	0.127
ENSG00000172531	PPP1CA*	0.087	0.066	0.121
ENSG00000163191	S100A11*	0.086	0.038	0.131
ENSG00000279483	*	0.086	0.060	0.108

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000170889	RPS9*	0.085	0.011	0.125
...ambiguous[ENSG00000241343+ENSG00000257529]	*	0.085	0.041	0.169
ENSG00000143947	RPS27A*	0.085	0.047	0.147
ENSG00000231925	TAPBP*	0.084	0.031	0.119
ENSG00000141741	MIEN1*	0.084	0.034	0.099
ENSG00000108344	PSMD3*	0.083	0.038	0.168
ENSG00000114353	GNAI2*	0.083	0.050	0.101
ENSG00000110651	CD81*	0.083	0.024	0.104
ENSG00000145425	RPS3A*	0.083	0.025	0.105
ENSG00000107223	EDF1*	0.082	0.052	0.109
ENSG00000171863	RPS7	0.081	-0.011	0.133
ENSG00000167526	RPL13*	0.079	0.055	0.190
ENSG00000184897	H1-10*	0.078	0.027	0.108
ENSG00000034510	TMSB10*	0.078	0.031	0.123
ENSG00000109475	RPL34*	0.078	0.056	0.169
ENSG00000167995	BEST1	0.078	-0.001	0.130
ENSG00000182809	CRIP2*	0.078	0.029	0.123
ENSG00000221983	UBA52*	0.077	0.049	0.122
ENSG00000071082	RPL31*	0.076	0.043	0.141
ENSG00000108639	SYNGR2*	0.076	0.038	0.139
...ambiguous[ENSG00000185883+ENSG00000260272]	*	0.076	0.048	0.108
ENSG00000131748	STARD3*	0.076	0.023	0.101
ENSG00000186468	RPS23*	0.076	0.025	0.130
ENSG00000174748	RPL15*	0.076	0.033	0.136
ENSG00000118181	RPS25*	0.076	0.028	0.137
ENSG00000105640	RPL18A*	0.073	0.050	0.124
ENSG00000102265	TIMP1*	0.072	0.031	0.120
ENSG00000124172	ATP5F1E*	0.070	0.034	0.195
ENSG00000083845	RPS5*	0.069	0.013	0.095
ENSG00000063177	RPL18*	0.069	0.034	0.081
ENSG00000130520	LSM4*	0.069	0.025	0.122
ENSG00000160789	LMNA*	0.069	0.034	0.117
ENSG00000105193	RPS16*	0.069	0.008	0.102
ENSG00000058262	SEC61A1*	0.069	0.049	0.114
ENSG00000008988	RPS20*	0.068	0.045	0.108
ENSG00000113140	SPARC*	0.068	0.023	0.113
ENSG00000130255	RPL36*	0.068	0.019	0.160
ENSG00000105372	RPS19*	0.068	0.026	0.164
ENSG00000196405	EVL	0.068	-0.006	0.147
ENSG00000138326	RPS24*	0.067	0.034	0.151
ENSG00000179218	CALR*	0.067	0.021	0.088
ENSG00000108518	PFN1*	0.067	0.039	0.110
ENSG00000142669	SH3BGRL3*	0.067	0.018	0.099
ENSG00000224389	C4B*	0.065	0.031	0.132

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000279274	*	0.065	0.017	0.100
ENSG00000025708	TYMP*	0.065	0.041	0.100
ENSG00000198034	RPS4X	0.064	-0.040	0.103
ENSG00000108679	LGALS3BP*	0.064	0.030	0.093
ENSG00000149806	FAU*	0.064	0.047	0.111
ENSG00000233927	RPS28*	0.064	0.012	0.087
ENSG00000143420	ENSA*	0.063	0.037	0.111
ENSG00000182899	RPL35A*	0.063	0.033	0.092
ENSG00000145592	RPL37*	0.063	0.025	0.117
ENSG00000173369	C1QB*	0.062	0.035	0.105
ENSG00000165215	CLDN3*	0.062	0.030	0.104
ENSG00000147403	RPL10*	0.062	0.019	0.089
ENSG00000164919	COX6C*	0.061	0.039	0.122
ENSG00000272196	*	0.061	0.032	0.107
ENSG00000142534	RPS11*	0.061	0.008	0.100
ENSG00000167468	GPX4*	0.061	0.046	0.082
ENSG00000051523	CYBA*	0.060	0.023	0.095
ENSG00000137154	RPS6*	0.060	0.031	0.123
ENSG00000182774	RPS17*	0.060	0.031	0.096
ENSG00000168528	SERINC2*	0.059	0.046	0.096
ENSG00000173801	JUP*	0.058	0.013	0.095
ENSG00000011304	PTBP1*	0.058	0.017	0.119
ENSG00000131469	RPL27	0.057	-0.008	0.083
ENSG00000106153	CHCHD2	0.057	-0.008	0.146
ENSG00000170315	UBB*	0.057	0.012	0.113
ENSG00000126267	COX6B1*	0.057	0.015	0.088
__ambiguous[ENSG00000219200+ENSG00000161939]	*	0.056	0.030	0.094
ENSG00000109062	NHERF1*	0.056	0.014	0.111
ENSG00000130303	BST2*	0.055	0.034	0.088
__ambiguous[ENSG00000254772+ENSG00000255508]		0.055	-0.011	0.074
ENSG00000125970	RALY*	0.055	0.009	0.138
ENSG00000155366	RHOC*	0.054	0.038	0.072
ENSG00000114867	EIF4G1*	0.054	0.022	0.074
ENSG00000156482	RPL30*	0.053	0.008	0.083
ENSG00000196924	FLNA*	0.053	0.043	0.095
ENSG00000142937	RPS8	0.053	-0.035	0.089
ENSG00000141753	IGFBP4	0.053	-0.005	0.107
ENSG00000185624	P4HB*	0.052	0.036	0.138
ENSG00000104964	TLE5*	0.052	0.030	0.064
ENSG00000104529	EEF1D	0.052	-0.019	0.067
ENSG00000172354	GNB2*	0.052	0.021	0.076
ENSG00000147684	NDUFB9*	0.050	0.020	0.082
ENSG00000134419	RPS15A	0.049	-0.002	0.165
ENSG00000132507	EIF5A	0.048	-0.013	0.070

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000150991	UBC	0.047	-0.013	0.089
ENSG00000142676	RPL11*	0.047	0.004	0.072
ENSG00000052344	PRSS8*	0.046	0.006	0.118
ENSG00000100316	RPL3*	0.046	0.026	0.110
ENSG00000100300	TSPO	0.046	-0.010	0.064
ENSG00000115268	RPS15*	0.045	0.010	0.066
ENSG00000168542	COL3A1	0.045	-0.014	0.253
ENSG00000104904	OAZ1	0.045	-0.007	0.086
ENSG00000174444	RPL4*	0.044	0.031	0.082
ENSG00000160932	LY6E*	0.044	0.032	0.092
ENSG00000213988	ZNF90*	0.042	0.015	0.074
ENSG00000117410	ATP6V0B	0.041	-0.015	0.070
ENSG00000114391	RPL24*	0.041	0.015	0.051
ENSG00000130725	UBE2M*	0.041	0.021	0.081
ENSG00000197958	RPL12	0.039	-0.014	0.072
..ambiguous[ENSG00000124614+ENSG00000270800]		0.039	-0.019	0.062
ENSG00000147604	RPL7	0.039	-0.021	0.091
..ambiguous[ENSG00000257341+ENSG00000213145]	*	0.038	0.005	0.068
ENSG00000172270	BSG*	0.037	0.021	0.049
ENSG00000167815	PRDX2*	0.037	0.019	0.062
..ambiguous[ENSG00000251357+ENSG00000240972]	*	0.036	0.005	0.073
ENSG00000167996	FTH1	0.035	-0.043	0.098
ENSG00000178719	GRINA	0.034	-0.001	0.086
ENSG00000186010	NDUFA13	0.034	-0.031	0.049
ENSG00000163479	SSR2	0.034	-0.008	0.049
ENSG00000135404	CD63	0.033	-0.016	0.063
ENSG00000092841	MYL6	0.033	-0.005	0.051
ENSG00000173272	MZT2A	0.032	-0.012	0.058
ENSG00000162244	RPL29	0.032	-0.004	0.075
ENSG00000178980	SELENOW*	0.032	0.004	0.071
ENSG00000105404	RABAC1*	0.032	0.005	0.080
ENSG00000172809	RPL38*	0.031	0.015	0.069
ENSG00000135480	KRT7	0.031	-0.001	0.096
ENSG00000140264	SERF2	0.031	-0.009	0.053
ENSG00000135940	COX5B	0.031	-0.016	0.071
ENSG00000124614	RPS10	0.030	-0.010	0.071
ENSG00000149925	ALDOA	0.030	-0.010	0.065
ENSG00000169223	LMAN2	0.030	-0.000	0.122
ENSG00000149273	RPS3	0.030	-0.010	0.090
ENSG00000115457	IGFBP2	0.029	-0.010	0.095
ENSG00000071553	ATP6AP1*	0.029	0.010	0.084
ENSG00000143774	GUK1	0.028	-0.004	0.072
ENSG00000111669	TPI1	0.028	-0.022	0.071
ENSG00000176046	NUPR1	0.026	-0.010	0.044

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000171858	RPS21	0.025	-0.015	0.050
ENSG00000214530	STARD10	0.022	-0.020	0.038
ENSG00000140988	RPS2	0.022	-0.002	0.039
ENSG00000136942	RPL35	0.022	-0.035	0.116
ENSG00000152082	MZT2B	0.022	-0.000	0.041
ENSG00000197956	S100A6	0.021	-0.023	0.063
ENSG00000123349	PFDN5*	0.020	0.009	0.052
ENSG00000127540	UQCR11	0.020	-0.023	0.035
ENSG00000204592	HLA-E*	0.019	0.006	0.128
...ambiguous[ENSG00000243678+ENSG00000011052]		0.019	-0.008	0.116
ENSG00000176340	COX8A*	0.019	0.005	0.045
ENSG00000188846	RPL14	0.018	-0.022	0.030
ENSG00000174886	NDUFA11*	0.018	0.011	0.033
ENSG00000167658	EEF2	0.018	-0.015	0.051
ENSG00000171222	SCAND1	0.017	-0.001	0.054
ENSG00000144426	NBEAL1	0.017	-0.009	0.049
ENSG00000148303	RPL7A	0.017	-0.016	0.056
ENSG00000169100	SLC25A6	0.016	-0.005	0.041
ENSG00000145494	NDUFS6	0.016	-0.008	0.045
ENSG00000111057	KRT18	0.016	-0.010	0.033
ENSG00000158710	TAGLN2*	0.016	0.002	0.049
...ambiguous[ENSG00000185885+ENSG00000185201]		0.015	-0.047	0.054
ENSG00000030582	GRN	0.015	-0.010	0.094
ENSG00000277791	PSMB3	0.014	-0.030	0.084
ENSG00000123989	CHPF	0.012	-0.011	0.091
ENSG00000131143	COX4I1	0.011	-0.016	0.022
ENSG00000198242	RPL23A	0.010	-0.037	0.045
ENSG00000161016	RPL8	0.010	-0.012	0.085
ENSG00000185825	BCAP31	0.009	-0.012	0.031
ENSG00000141522	ARHGDIA	0.008	-0.015	0.038
ENSG00000277972	CISD3	0.008	-0.013	0.043
...ambiguous[ENSG00000262526+ENSG00000170296]		0.007	-0.043	0.051
ENSG00000185499	MUC1	0.005	-0.076	0.029
ENSG00000074800	ENO1	0.005	-0.052	0.051
ENSG00000172757	CFL1	0.004	-0.020	0.029
ENSG00000177700	POLR2L	0.003	-0.025	0.037
ENSG00000178952	TUFM	0.003	-0.019	0.068
ENSG00000165949	IFI27	0.001	-0.019	0.017
ENSG00000163220	S100A9	0.001	-0.028	0.050
ENSG00000187608	ISG15	0.001	-0.021	0.067
ENSG00000126247	CAPNS1	0.000	-0.022	0.013
ENSG00000111341	MGP	-0.002	-0.099	0.050
ENSG00000110492	MDK	-0.002	-0.046	0.038
ENSG00000135390	ATP5MC2	-0.002	-0.056	0.038

continued

Gene	Symbol	median r	CI lo	CI hi
__ambiguous[ENSG00000273259+ENSG00000196136]		-0.004	-0.035	0.018
ENSG00000140990	NDUFB10	-0.004	-0.019	0.023
ENSG00000103363	ELOB	-0.005	-0.029	0.034
ENSG00000144713	RPL32	-0.006	-0.047	0.023
ENSG00000184840	TMED9	-0.006	-0.093	0.035
ENSG00000143546	S100A8	-0.006	-0.054	0.047
ENSG00000204628	RACK1	-0.008	-0.020	0.046
ENSG00000142089	IFITM3	-0.011	-0.041	0.046
ENSG00000160862	AZGP1	-0.011	-0.066	0.005
ENSG00000246705	H2AJ	-0.012	-0.040	0.039
ENSG00000159335	PTMS	-0.020	-0.030	0.027
ENSG00000160180	TFF3	-0.023	-0.079	0.013
ENSG00000125691	RPL23	-0.043	-0.099	0.004
ENSG00000110092	CCND1	-0.046	-0.119	0.054
ENSG00000125534	PPDPF	-0.050	-0.071	0.007
ENSG00000108298	RPL19	-0.065	-0.137	-0.012

C Locked formulas index

1. Pearson $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$
2. Spearman ρ (Pearson on ranks)
3. Ridge $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$
4. Gaussian kernel $W_{ij} = \exp(-d_{ij}^2/2\ell^2)$, row-normalized
5. Spatial smoothing $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W \hat{y}$
6. Moran's I (see Methods)
7. Beer–Lambert $OD = -\ln(I/I_0)$; deconvolution $C = ODM^{-1}$
8. GLCM contrast / homogeneity / energy / correlation (see Methods)
9. Shannon entropy $H = -\sum p \log_2 p$
10. AUC via Mann–Whitney $U/(n_1 n_0)$
11. Benjamini–Hochberg adjusted $p_{(i)} \cdot m/i$
12. Permutation $p = (1 + \#\{|null| \geq |obs|\})/(1 + n)$
13. Wilcoxon signed-rank statistic
14. Brier score $\frac{1}{n} \sum (p_i - y_i)^2$
15. Softmax $\alpha_i = e^{s_i/\tau} / \sum_j e^{s_j/\tau}$
16. Library normalization $\log(1 + 10^4 x / \sum x)$

Provenance note. The appendix tables from here on (per-fold, transport, scanner coefficients, top predictable genes, enrichment, ledgers, cohort manifest) are Phase-1 artifacts from the handcrafted-feature pipeline (commit 84b076a), preserved for provenance. Phase-2 equivalents are Appendix 5–6 and the refreshed scanner outputs in `results/`.

D Per-fold results

Patient	sections	λ	α	M1 median r	M2 median r
BC23287	3	100.0	0.5	-0.007	-0.010
BC23450	3	100.0	0.4	0.016	0.019
BC23567	3	100.0	0.5	0.009	0.009
BC23810	3	100.0	0.5	0.011	0.014
BC23901	2	100.0	0.5	0.009	0.013
BC23903	3	100.0	0.5	0.043	0.055
BC23944	3	100.0	0.5	0.029	0.033
BC24044	3	100.0	0.5	0.040	0.049
BC24105	3	100.0	0.5	-0.017	-0.022
BC24220	3	100.0	0.5	0.025	0.034

E Transport results

Subtype	Section	median per-gene r
Luminal_A	BC23268_C1	0.072
Luminal_A	BC23268_C2	0.061
Luminal_A	BC23268_D1	0.076
Luminal_A	BC23269_C1	0.040
Luminal_A	BC23269_C2	0.025
Luminal_A	BC23269_D1	0.031
Luminal_A	BC23272_D2	0.050
Luminal_A	BC23272_E1	0.075
Luminal_A	BC23272_E2	0.021
Luminal_A	BC24223_D2	-0.024
Luminal_A	BC24223_E1	-0.013
Luminal_A	BC24223_E2	0.000
Luminal_B	BC23270_D2	-0.007
Luminal_B	BC23270_E1	0.041
Luminal_B	BC23270_E2	0.023
Luminal_B	BC23277_D2	0.015
Luminal_B	BC23277_E1	0.027
Luminal_B	BC23277_E2	-0.025
Luminal_B	BC23506_C1	0.016
Luminal_B	BC23506_C2	-0.004

Subtype	Section	median r
Luminal_B	BC23506_D1	-0.001
Luminal_B	BC23508_D2	0.037
Luminal_B	BC23508_E1	0.026
Luminal_B	BC23508_E2	0.043
Luminal_B	BC23895_C1	0.013
Luminal_B	BC23895_C2	0.014
Luminal_B	BC23895_D1	0.005
TNBC	BC23209_C1	0.004
TNBC	BC23209_C2	0.004
TNBC	BC23209_D1	-0.024
TNBC	BC23288_D2	0.021
TNBC	BC23288_E1	0.051
TNBC	BC23288_E2	0.029
TNBC	BC23377_C1	0.010
TNBC	BC23377_C2	-0.005
TNBC	BC23377_D1	-0.007
TNBC	BC23803_D2	0.006
TNBC	BC23803_E1	0.038
TNBC	BC23803_E2	0.019

F Scanner coefficients

Feature	coefficient	z	p
glcm0_energy	24.566	5.52	3.41e-08
glcm1_energy	-23.145	-5.13	2.90e-07
glcm3_energy	12.963	3.56	3.67e-04
glcm2_energy	-12.931	-3.60	3.23e-04
b_mean	11.592	18.82	5.46e-79
r_mean	-5.979	-19.25	1.51e-82
eosin_mean	5.229	12.23	2.26e-34
dab_mean	4.647	15.01	6.58e-51
r_std	2.601	12.53	5.15e-36
g_mean	-2.417	-7.49	6.93e-14
hema_mean	2.409	10.49	9.52e-26
hema_std	-2.103	-17.03	4.88e-65
eosin_std	2.020	12.40	2.58e-35
b_std	-1.372	-8.12	4.63e-16
glcm2_homog	-1.319	-1.50	1.33e-01
g_std	-1.137	-5.40	6.84e-08
glcm1_homog	-0.915	-1.10	2.69e-01
gray_entropy	-0.745	-9.48	2.61e-21
glcm3_homog	0.649	0.75	4.55e-01
glcm0_homog	-0.617	-0.74	4.60e-01

Feature	coef	z	p
glcm1_contrast	0.528	0.96	3.35e-01
r_skew	-0.515	-3.55	3.84e-04
hema_entropy	0.422	11.23	2.77e-29
glcm3_corr	0.387	4.08	4.59e-05
glcm1_corr	-0.365	-4.66	3.18e-06
grad_mag_mean	-0.365	-2.00	4.55e-02
dab_std	0.279	3.92	8.80e-05
glcm2_contrast	-0.247	-0.43	6.66e-01
g_skew	0.236	1.53	1.25e-01
glcm0_corr	-0.213	-2.69	7.05e-03
glcm0_contrast	0.189	0.34	7.34e-01
glcm2_corr	-0.135	-1.44	1.51e-01
hema_dense_frac	0.133	5.05	4.52e-07
r_kurt	-0.114	-0.76	4.48e-01
hema_fracpos	0.078	3.38	7.22e-04
b_kurt	-0.065	-0.42	6.77e-01
dab_fracpos	0.045	2.01	4.47e-02
glcm3_contrast	0.019	0.03	9.74e-01
b_skew	0.015	0.10	9.19e-01
eosin_fracpos	-0.006	-0.24	8.09e-01
g_kurt	0.001	0.01	9.95e-01

G Top predictable genes

Gene (Ensembl)	M1 median r	M2 median r
ENSG00000177600	0.144	0.154
ENSG00000096384	0.140	0.138
ENSG00000141738	0.115	0.140
ENSG00000137818	0.111	0.144
ENSG00000269028	0.110	0.153
ENSG00000108821	0.109	0.137
ENSG00000132475	0.105	0.140
ENSG00000255823	0.104	0.122
ENSG00000166710	0.098	0.114
ENSG00000087086	0.090	0.104
ENSG00000196576	0.087	0.089
ENSG00000279483	0.087	0.104
ENSG00000130203	0.086	0.120
ENSG00000161960	0.078	0.116
ENSG00000168528	0.076	0.095
ENSG00000115414	0.075	0.082
ENSG00000169710	0.073	0.090
ENSG00000141736	0.068	0.077

Gene	M1	M2
ENSG00000087460	0.068	0.092
ENSG00000234745	0.067	0.080
ENSG00000058262	0.064	0.082
ENSG00000102265	0.063	0.054
ENSG00000204287	0.061	0.079
ENSG00000177954	0.061	0.070
ENSG00000075624	0.061	0.085
ENSG00000108344	0.060	0.052
ENSG00000170421	0.059	0.061
ENSG00000089157	0.056	0.058
ENSG00000108518	0.055	0.070
ENSG00000132507	0.055	0.061
ENSG00000130255	0.054	0.073
ENSG00000205542	0.053	0.069
ENSG00000164692	0.053	0.065
ENSG00000197746	0.053	0.069
ENSG00000026025	0.052	0.069
ENSG00000083845	0.051	0.051
ENSG00000100097	0.051	0.055
ENSG00000182899	0.050	0.050
ENSG00000167526	0.049	0.045
ENSG00000142937	0.048	0.046
ENSG00000130725	0.048	0.047
ENSG00000163399	0.047	0.069
ENSG00000164733	0.047	0.064
ENSG00000197756	0.046	0.060
ENSG00000100219	0.046	0.047
ENSG00000108107	0.046	0.050
ENSG00000108639	0.046	0.038
ENSG00000105223	0.046	0.052
__ambiguous [ENSG00000277957+ENSG00000161960]	0.045	0.071
ENSG00000134419	0.044	0.065
ENSG00000019582	0.043	0.038
ENSG00000149806	0.043	0.037
ENSG00000159840	0.043	-0.006
ENSG00000142949	0.043	0.053
ENSG00000198034	0.042	0.031
ENSG00000184009	0.041	0.042
ENSG00000105193	0.041	0.047
__ambiguous [ENSG00000185883+ENSG00000260272]	0.041	0.050
ENSG00000125970	0.040	0.069
ENSG00000164587	0.040	0.049
ENSG00000158710	0.040	0.031
ENSG00000125730	0.039	0.043
ENSG00000123416	0.038	0.044

Gene	M1	M2
ENSG00000173369	0.038	0.034
ENSG00000142173	0.037	0.049
ENSG00000114867	0.037	0.045
ENSG00000173272	0.035	0.023
ENSG00000117450	0.035	0.062
ENSG00000279274	0.035	0.054
ENSG00000229117	0.035	0.026
ENSG00000231500	0.034	0.047
ENSG00000130520	0.034	0.031
ENSG00000106153	0.034	0.023
ENSG00000142541	0.034	0.041
ENSG00000163191	0.034	0.051
ENSG00000231925	0.033	0.037
ENSG00000145425	0.033	0.050
ENSG00000104904	0.033	0.038
ENSG00000131748	0.033	0.056
ENSG00000105404	0.032	0.028
__ambiguous [ENSG00000251357+ENSG00000240972]	0.032	0.024
ENSG00000196405	0.032	0.032
ENSG00000126267	0.031	0.039
ENSG00000174444	0.030	0.023
ENSG00000185624	0.030	0.028
ENSG00000118181	0.029	0.030
ENSG00000111640	0.029	0.033
ENSG00000187514	0.028	0.056
ENSG00000170315	0.028	0.015
ENSG00000224389	0.028	0.019
ENSG00000233927	0.028	0.022
ENSG00000143420	0.027	0.037
ENSG00000213741	0.027	0.023
ENSG00000051523	0.027	0.008
ENSG00000277791	0.027	0.042
ENSG00000071082	0.027	0.051
ENSG00000128322	0.027	0.003
ENSG00000071553	0.026	0.038
ENSG00000277972	0.026	0.016
ENSG00000163682	0.025	0.043
ENSG00000109062	0.025	0.040
ENSG00000272196	0.025	0.025
ENSG00000034510	0.024	0.014
ENSG00000123989	0.024	0.020
ENSG00000113140	0.024	0.022
__ambiguous [ENSG00000243678+ENSG00000011052]	0.023	0.029
ENSG00000171863	0.023	0.042
ENSG00000140264	0.023	0.034

Gene	M1	M2
ENSG00000167468	0.022	0.026
ENSG00000178952	0.022	0.017
...ambiguous [ENSG00000241343+ENSG00000257529]	0.022	0.024
ENSG00000204525	0.022	0.056
ENSG00000141741	0.022	0.010
ENSG00000170889	0.021	0.019
ENSG00000141522	0.021	0.025
ENSG00000142676	0.021	0.015
ENSG00000140988	0.021	0.020
ENSG00000110651	0.020	0.026
ENSG00000115457	0.020	0.004
ENSG00000162244	0.020	0.015

H Dataset ledger composition

The 629-record accession ledger summarized by source, with example accessions; the full ledger is `results/accession_ledger.csv`.

Source	Records	Level	Example accessions
CELLxGENE	40	metadata-mined	cxg:af893e86-8e9f-41f1-a474-ef05359b1fb7; cxg:3a5dbf8a-9b3e-4309-b4c5-d8a024f83734; cxg:16876983-d454-43db-a8ec-77fafca9bf38
GDC	80	metadata-mined	TCGA-LUAD; ALCHEMIST-ALCH; MATCH-S1
GEO-GSE	400	metadata-mined	GSE344506; GSE348445; GSE337859
Mendeley-29ntw7sh4r-v5	69	analyzed-primary	BC23287_C1; BC23287_C2; BC23287_D1
TCIA	40	metadata-mined	tcia:4D-Lung; tcia:A091105; tcia:ACNS0332
Total	629		

I Cohort manifest

One row per analyzed section. Tumor fraction is the share of spots under the pathologist tumor annotation for that section.

Patient	Section	Subtype	Spots	Tumor frac.	HER2+
BC23287	BC23287_C1	HER2_luminal	256	0.00	yes
BC23287	BC23287_C2	HER2_luminal	334	0.00	yes
BC23287	BC23287_D1	HER2_luminal	357	0.00	yes
BC23450	BC23450_D2	HER2_luminal	316	0.00	yes
BC23450	BC23450_E1	HER2_luminal	301	0.00	yes
BC23450	BC23450_E2	HER2_luminal	318	0.00	yes
BC23901	BC23901_C2	HER2_luminal	292	0.00	yes

Patient	Section	Subtype	Spots	Tumor	HER2+
BC23901	BC23901_D1	HER2_luminal	297	0.00	yes
BC23944	BC23944_D2	HER2_luminal	414	0.00	yes
BC23944	BC23944_E1	HER2_luminal	423	0.00	yes
BC23944	BC23944_E2	HER2_luminal	403	0.00	yes
BC24220	BC24220_D2	HER2_luminal	432	0.00	yes
BC24220	BC24220_E1	HER2_luminal	428	0.00	yes
BC24220	BC24220_E2	HER2_luminal	437	0.00	yes
BC23567	BC23567_D2	HER2_non_luminal	587	0.00	yes
BC23567	BC23567_E1	HER2_non_luminal	572	0.00	yes
BC23567	BC23567_E2	HER2_non_luminal	570	0.00	yes
BC23810	BC23810_D2	HER2_non_luminal	691	0.00	yes
BC23810	BC23810_E1	HER2_non_luminal	695	0.00	yes
BC23810	BC23810_E2	HER2_non_luminal	713	0.00	yes
BC23903	BC23903_C1	HER2_non_luminal	441	0.00	yes
BC23903	BC23903_C2	HER2_non_luminal	468	0.00	yes
BC23903	BC23903_D1	HER2_non_luminal	464	0.00	yes
BC24044	BC24044_D2	HER2_non_luminal	614	0.00	yes
BC24044	BC24044_E1	HER2_non_luminal	603	0.00	yes
BC24044	BC24044_E2	HER2_non_luminal	511	0.00	yes
BC24105	BC24105_C1	HER2_non_luminal	325	0.00	yes
BC24105	BC24105_C2	HER2_non_luminal	297	0.00	yes
BC24105	BC24105_D1	HER2_non_luminal	338	0.00	yes
BC23268	BC23268_C1	Luminal_A	435	0.00	no
BC23268	BC23268_C2	Luminal_A	498	0.00	no
BC23268	BC23268_D1	Luminal_A	492	0.00	no
BC23269	BC23269_C1	Luminal_A	434	0.00	no
BC23269	BC23269_C2	Luminal_A	435	0.00	no
BC23269	BC23269_D1	Luminal_A	433	0.00	no
BC23272	BC23272_D2	Luminal_A	565	0.00	no
BC23272	BC23272_E1	Luminal_A	578	0.00	no
BC23272	BC23272_E2	Luminal_A	548	0.00	no
BC24223	BC24223_D2	Luminal_A	368	0.00	no
BC24223	BC24223_E1	Luminal_A	362	0.00	no
BC24223	BC24223_E2	Luminal_A	363	0.00	no
BC23270	BC23270_D2	Luminal_B	284	0.00	no
BC23270	BC23270_E1	Luminal_B	283	0.00	no
BC23270	BC23270_E2	Luminal_B	284	0.00	no
BC23277	BC23277_D2	Luminal_B	506	0.00	no
BC23277	BC23277_E1	Luminal_B	513	0.00	no
BC23277	BC23277_E2	Luminal_B	566	0.00	no
BC23506	BC23506_C1	Luminal_B	489	0.00	no
BC23506	BC23506_C2	Luminal_B	502	0.00	no
BC23506	BC23506_D1	Luminal_B	482	0.00	no
BC23508	BC23508_D2	Luminal_B	478	0.00	no
BC23508	BC23508_E1	Luminal_B	473	0.00	no

Patient	Section	Subtype	Spots	Tumor	HER2+
BC23508	BC23508_E2	Luminal_B	504	0.00	no
BC23895	BC23895_C1	Luminal_B	498	0.00	no
BC23895	BC23895_C2	Luminal_B	454	0.00	no
BC23895	BC23895_D1	Luminal_B	518	0.00	no
BC23209	BC23209_C1	TNBC	294	0.00	no
BC23209	BC23209_C2	TNBC	318	0.00	no
BC23209	BC23209_D1	TNBC	331	0.00	no
BC23288	BC23288_D2	TNBC	464	0.00	no
BC23288	BC23288_E1	TNBC	480	0.00	no
BC23288	BC23288_E2	TNBC	478	0.00	no
BC23377	BC23377_C1	TNBC	597	0.00	no
BC23377	BC23377_C2	TNBC	687	0.00	no
BC23377	BC23377_D1	TNBC	601	0.00	no
BC23803	BC23803_D2	TNBC	402	0.00	no
BC23803	BC23803_E1	TNBC	340	0.00	no
BC23803	BC23803_E2	TNBC	421	0.00	no

J Pathway enrichment of the top-25 predictable genes

g:Profiler g:GOST (GO:BP, KEGG, Reactome) on the 25 genes with the highest held-out M1 correlation; full response in results/enrichment_top25.json.

Source	Term	<i>p</i>	Genes
KEGG	Antigen processing and presentation	9.22e-04	23
REAC	Scavenging by Class A Receptors	1.82e-03	23
GO:BP	antigen processing and presentation of endogenous peptide antigen	3.12e-03	23
REAC	GRB7 events in ERBB2 signaling	6.68e-03	23
GO:BP	antigen processing and presentation of endogenous antigen	6.90e-03	23
GO:BP	wound healing	7.13e-03	23
GO:BP	protein biosynthetic process	8.61e-03	23
GO:BP	translation	8.61e-03	23
REAC	Disease	9.90e-03	23
KEGG	Vibrio cholerae infection	1.04e-02	23
GO:BP	positive regulation of T cell mediated cytotoxicity	1.19e-02	23
REAC	Axon guidance	1.37e-02	23
GO:BP	regulation of cell adhesion	1.57e-02	23
GO:BP	regulation of developmental process	1.61e-02	23
KEGG	Phagosome	1.68e-02	23

K Test inventory

Test module	Tests
features	11
metrics	21
Total	32

The 32 unit tests pin every metric against known values (perfect/anti correlation, constant-input degeneracies, ridge recovery and shrinkage, kernel normalization, smoothing identity at $\alpha = 0$, Moran's I sign on clustered vs alternating patterns, AUC 0/0.5/1 anchors, BH monotonicity, permutation- p bounds, Brier anchors, softmax normalization and temperature behavior) and every feature extractor against physical expectations (white-field OD ≈ 0 , constant-image GLCM energy 1 and contrast 0, uniform-histogram entropy maximum, feature-vector finiteness and name uniqueness, textured-vs-flat discriminability).